MILITARY SPECIFICATION

TRANSMITTING SET, RADAR DATA AN/AKT-18B
AND
RECEIVING SET, RADAR DATA AN/TKQ-2B (SYSTEM OF)

This specification is approved for use by Electronics Research and Development Command, Department of the Army, and is available for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the following subsystems and units of the data link for Radar Surveillance System AN/UPD-7:

Transmitting Set, Radar Data AN/AKT-18B (airborne systems)
Receiving Set, Radar Data AN/TKQ-2B (ground station)

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The effective issue or revision of the following documents shall be that listed in the Department of Defense Index of Specifications and Standards (DODISS) and supplements thereto, unless (i) specific issues are set forth therefor in the cited specifications or (ii) issues different from those specified in the cited specifications are set forth in the solicitation. The date of the applicable DODISS and supplements thereto shall be as specified in the solicitation or contract.

SPECIFICATIONS

MILITARY

MIL-P-116	- Preservation-Packaging, Methods of
MIL-D-49375(ER)	- Decoder, Video KY-871( )/TKQ-2B
MIL-E-49387(ER)	- Encoder, Video KY-865I( )/AKT-18B and Control, Data Link C-10546( )/AKT-18B
MIL-S-55507	- Shelter, Electrical Equipment (with or without equipment), Packaging of

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Electronics Research and Development Command, ATTN: DELCS-PE, Fort Monmouth, New Jersey 07703 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-252	- Classification of Visual and Mechanical Defects for Equipment, Electronc, Wired and Other Devices.
MIL-STD-454	- Standard General Requirements for Electronic Equipment.
MIL-STD-1472	- Human Engineering Design Criteria for Military System, Equipment and Facilities.

DRAWINGS

SM-D-942276	- Transmitting Set, Radar Data AN/AKT-18B.
SM-D-942277	- Receiving Set, Radar Data AN/TKQ-2B.
SM-A-942315	- Performance Test Under Laboratory Conditions for Receiving Set, Radar Data AN/TKQ-2B P/N SM-D-942277.
SM-A-942912	- Test Procedure for Digital Data Link Airborne Subsystem Test AN/AKT-18B.
SM-A-942916	- Test Procedure for Digital Data Link Ground Subsystem Test AN/TKQ-2B.
SM-A-942941	- Test Procedure for SLAR Digital Data Link System Bench Test AN/AKT-18B and AN/TKQ-2B.
SM-A-946269	- Flight Test Prodedure for AN/APS-94F to Digital Data Link Compatibility.
SPI IG00248	- Special Packaging AN/AKT-18B.

2.2 OTHER PUBLICATIONS

Title 29 Code of Federal Regulations, Chapter XVII, Part 1910.

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 First article. When specified (see 6.2) the contractor shall furnish first article units for first article inspection and approval (see 4.3).

3.2 Construction. The Transmitting Set, Radar Data AN/AKT-18B and Receiving Set, Radar Data AN/TKQ-2B, shall be constructed in accordance with the requirements of this specification and the following drawings and specifications:

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a. Transmitting Set: SM-D-942276
MIL-E-49387(ER) (Encoder)

b. Receiving Set: SM-D-942277
MIL-D-49375(ER) (Decoder)

3.3 Parts, materials, and processes; general. In addition to the requirements of this specification, the requirements for each unit of AN/AKT-18B and AN/TKQ-2B as specified in its specification or drawings, should apply (see 4.4).

3.4 Recycled, virgin and reclaimed materials. It is required that, to the maximum extent possible, recovered or reclaimed materials, in lieu of virgin materials, shall be used without jeopardizing the intended use of the manufactured item.

3.5 Finish. All surfaces of each unit of AN/AKT-18B and AN/TKQ-2B shall be finished as specified in its unit specifications and drawings (see 4.4).

3.6 Nameplate and marking.

3.6.1 General. Each unit of AN/AKT-18B and AN/TKQ-2B shall be marked, nomenclatured, and serial-numbered as specified in its unit specifications and drawings (see 4.4).

3.7 Unit safety engineering. The requirements of safety engineering shall apply to each of unit of AN/AKT-18B and AN/TKQ-2B as specified in its unit specifications and drawings (see 4.13).

3.7.1 System safety engineering. Each unit shall minimize the probability and severity of injury to personnel throughout the systems life cycle and be in accordance with Title 29 Code of Federal Regulations, Chapter XVII, Part 1910, Occupational Safety and Health Standards, Requirement 1 MIL-STD-454 and shall include but not be limited to paragraphs entitled Labeling, Hazards and Safety of MIL-STD-1472.

3.8 Interchangeability. Like units, assemblies, subassemblies and replaceable parts shall conform to Requirement 7 of MIL-STD-454 (see 4.8).

3.9 Weight. The weight of the system shall not exceed the sum of weights of the units of AN/AKT-18B and AN/TKQ-2B as specified in unit specifications, excluding cabling (see 4.7).

3.10 Burn-in. Each unit of AN/AKT-18B and AN/TKQ-2B shall be subjected to the burn-in cycle as specified in its specification (see 4.6).

3.11 Performance requirements.

3.11.1 Transmitter Set, Radar Data AN/AKT-18B. The AN/AKT-18B shall meet the subsystem performance requirements after each unit has met the following requirements:

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Unit	Requirement
Encoder, Video KY-865( )/AKT-18B	MIL-E-49387(ER)
Control, Data Link C-10546( )/AKT-18B	MIL-E-49387(ER)
Antenna AT-450( )/ARC	GFE
Mounting Base, Electrical Equipment MT-6016( )/AKT-18B	SM-D-942282
Radio Set AN/ARC-164(V)16	SM-D-943231

3.11.1.1 AN/AKT-18B subsystem requirements. The AN/AKT-18B shall be tested in accordance with and meet the requirements of SM-A-942912 (see 4.9.1).

3.11.2 Receiving Set, Radar Data AN/TKQ-2B. The AN/TKQ-2B shall meet the subsystem performance requirements below, after each unit has met the following requirements:

Unit	Requirement
Decoder, Video KY-871( )/TKQ-2B	MIL-D-49375(ER)
Shelter, Electrical Equipment S-606( )/TKQ-2B	SM-D-942283
Antenna-Radome AS-1097( )/GR	GFE
or	
Antenna AT-197( )/GR w/Adapter, Antenna to Antenna Base MX-10060( )/TKQ-2B	SM-D-942284
Mast AB-924( )/TKQ-2	SM-D-579043
Radio Set AN/ARC-164(V)16	SM-D-943231
Power Supply PP-6305( )/G	GFE
Mounting Base, Electrical Equipment MT-6052( )/TKQ-2B	SM-D-942286
Recorder-Processor-Viewer, Radar Mapping RO-495( )/U	SM-D-878155
Case, Storage CY-7760( )/U	SM-D-942858
Antenna AT-912( )/VRC w/Antenna Matching Unit MX-2799( )/VRC	GFE
or	
Antenna AS-1729( )/VRC w/Antenna Matching Unit MX-6706( )/VRC	
Accessory Kit, Data Link MK-1951( )/TKQ-2B	SM-D-942831
Receiving-Transmitter, Radio RT-524( )/VRC	GFE
Mounting Base, Electrical Equipment MT-1029( )/VRC	GFE
Oscilloscope AN/USM-281C	GFE

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3.11.2.1 AN/TKQ-2B subsystem requirements. The AN/TKQ-2B shall be tested in accordance with and meet the requirements of SM-A-942916 and SM-A-942315 (see 4.9.2).

3.11.3 AN/AKT-18B and AN/TKQ-2B data link system performance. The AN/AKT-18B and AN/TKQ-2B shall meet the requirements of SM-A-942941 Test Procedure for SLAR Digital Data Link System Bench Test AN/AKT-18B and AN/TKQ-2B and SM-A-946269 Flight Test Procedure for AN/APS-94F to Digital Data Link Compatibility (see 4.9.3).

3.12 Service conditions.

3.12.1 Transmitter Set, Radio Data AN/AKT-18B. Each unit of the transmitter set shall meet the environmental conditions as required on the specification or drawing as specified in 3.11.1.

3.12.2 Receiving Set, Radar Data AN/TKQ-2B. Each unit of the receiving set shall meet the environmental conditions as required on the specification or drawing as specified in 3.11.2.

3.12.3 AN/AKT-18B and AN/TKQ-2B Data Link System. AN/AKT-18B and AN/TKQ-2 data link system shall be submitted to the environment condition as follows:

a. Room temperature during bench test
b. Natural flight environment condition (see 3.11.3)

3.13 Electromagnetic interference. The AN/AKT-18B and AN/TKQ-2B equipments as installed in the aircraft and shelter shall not malfunction, exhibit degraded performance or undesirable response when electronics associated with the system are singularly or collectively operated. Likewise, the AN/AKT-18B and AN/TKQ-2B equipments shall not cause a malfunction, degraded performance or undesirable response in any other equipment or subsystem associated with the system when performing their intended transmitting and receiving functions.

3.14 Reliability. Each unit of AN/AKT-18B and AN/TKQ-2B shall meet the specified reliability as required in the specification or drawings.

3.15 Workmanship. The AN/AKT-18B and AN/TKQ-2B shall be manufactured and assembled in accordance with Requirement 9 of MIL-STD-454.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the government. The government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3). (Does not include packaging.)

b. Inspections covered by subsidiary documents (see 4.4).

c. Quality conformance inspections.

Quality conformance inspection of equipment before packaging (see 4.5).
Quality conformance inspection of packaging (see 4.14).

4.3 First article. Unless otherwise specified in the contract, the first article inspection shall be performed by the contractor.

4.3.1 First article units. The contractor shall furnish two first article systems of the AN/AKT-18B and AN/TKQ-2B for group A testing.

4.3.2 First article inspection. The first article inspection shall consist of the inspections specified in table I, and shall be performed in the order specified.

4.3.3 First article data. The first article test plan and test reports shall be as required in the contract.

TABLE I. First article inspection.

Inspections	Reqt para	Test para	Order of tests One system
1. Inspections covered by subsidiary documents 1/	3.3 3.5 3.6 3.15	4.4	Inspection to be performed on all units
2. Group A inspection 1/	See table II		Tests to be performed on all units
3. Group B inspection 1/	See table III		
4. Group C inspection EMI Flight test	 3.13 3.11.3	 4.11 4.9.3	

1/ The inspection 1, 2 and 3 in the order shown, shall be performed on all first article units before subjecting these units to any other inspection requirements.

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4.4 Inspections covered by subsidiary documents. The following shall be inspected under the applicable documents as part of the inspection required by this specification, and the inspection requirement specified in the contract.

Item	Where required
Parts, materials, and processes; general	3.3
Finish	3.5
Nameplate and marking	3.6

4.5 Quality conformance inspection of equipment before packaging. The contractor shall perform the inspections specified in 4.4, 4.5.1 through 4.5.3. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements.

4.5.1 Group A inspection. Each system on contract shall be inspected for conformance to the inspection in table II. Discrete lots shall be formed from units that pass this inspection. Factors of lot composition not defined herein or in the contract, shall be in accordance with MIL-STD-105. Each lot shall be subject to sampling inspection, utilizing the procedures of MIL-STD-105, using the general inspection levels, and AQL's indicated in table II.

4.5.1.1 Order of inspection within group A. Group A inspection shall be performed in an order satisfactory to the government.

TABLE II. Group A inspection.

Inspection	Requirement paragraph	Test paragraph	AQL major	AQL minor
Visual and mechanical	3.15	4.13	1%	4%
System safety	3.7	4.13	1%	4%
EMI 1/	3.13	4.12		
Performance				
AN/AKT-18B subsystem	3.11.1.1	4.9.1		2/
AN/AKQ-2B subsystem	3.11.2.1	4.9.2		2/
AN/AKT-18B and AN/AKQ-2B system:				
Bench test	3.11.3	4.9.3		2/
Flight Test 1/	3.11.3	4.9.3		2/

1/ These inspections shall be performed once at first production system.
2/ All electrical defects are major.

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4.5.2 Group B inspection. Group B inspection shall normally be performed in inspection lots that have passed group A inspection and on samples selected from units that have been subjected to and met the group A inspection. This inspection shall conform to table III and to the special inspection levels of table I of MIL-STD-105.

4.5.2.1 Group B sampling plan. The group B sampling plans, for the AQL's listed in table III shall be as follows:

AQL	Inspection level
4.0 percent	S-1
6.5 percent	S-1

4.5.2.2 Order of inspection within group B. Group B inspection shall be performed in an order satisfactory to the government.

TABLE III. Group B inspection.

Inspection	Requirement paragraph	Test paragraph	AQL
Interchangeability			
Mechanical	3.8	4.8	6.5%
Electrical	3.8	4.8	4%

4.5.3 Group C inspection. Group C inspection shall be performed on units that have passed group A and B inspection. The inspection shall consist of the inspection specified in table IV. Samples shall be selected in accordance with 4.5.3.1.

4.5.3.1 Sampling for group C inspection. One sample system shall be selected at random for each inspection in table IV. The first samples selected shall be at the start of the contract from the first quality conformance inspection production lot. These samples shall constitute the group C requirement for all units produced.

4.5.3.2 Group C failures. Action required relative to group C failure shall be as specified in the contract.

4.5.3.3 Reinspection of conforming group C sample units. Unless otherwise specified, sample units which have been subjected to and passed group C inspection may be accepted on the contract provided all damage is repaired and the sample units are resubjected to and pass group A inspection only.

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TABLE IV. Group C inspection.

Inspection	Requirement paragraph	Test paragraph
EMI	3.13	4.11
Flight Test	3.11.3	4.9.3

4.6 Burn-in. Each unit of AN/AKT-18B and AN/TKQ-2B shall meet the requirement as specified in its specification prior to being subjected to the system test specified by 4.9 (see 3.10).

4.7 Weight. The weight of the system shall not exceed the sum of weights of the units of AN/AKT-18B and AN/TKQ-2B, excluding cabling (see 3.9).

4.8 Interchangeability. The mechanical dimensions and electrical performance of parts shall be physically and electrically interchangeable among the units. Mechanical dimensions and electrical performance shall be measured to determine compliance with Requirement 7 of MIL-STD-454 (see 3.8).

4.9 Performance testing.

4.9.1 AN/AKT-18B subsystem performance testing. AN/AKT-18B subsystem shall be tested for compliance with specified requirements of SM-A-942912 (see 3.11.1.1).

4.9.2 AN/TKQ-2B subsystem performance testing. AN/TKQ-2B subsystem shall be tested for compliance with specified requirements of SM-A-942916 and SM-A-942315 (see 3.11.2.1).

4.9.3 AN/AKT-18B and AN/TKQ-2B data link system performance testing. AN/AKT-18B and AN/TKQ-2B data link system shall be tested for compliance with specified performance requirements of SM-A-942941, Test Procedure for SLAT Digital Data Link System Bench Test AN/AKT-18B and AN/TKQ-2B and SM-A-946269, Flight Test Procedure for AN/APS-94F to Digital Data Link Compatibility (see 3.11.3).

4.10 Service conditions. The service condition tests shall be performed as detailed in 3.12.

4.11 Electromagnetic interference. The contractor shall devise and implement test procedures to demonstrate intrasystem electromagnetic compatibility of the equipments as installed in the shelter and the aircraft (see 3.13).

4.12 Reliability tests. Each unit of AN/AKT-18B and AN/TKQ-2B shall meet the requirement as specified in its specification prior to being subjected to the system test specified by 4.9 (see 3.14).

4.13 Visual and mechanical. The AN/AKT-18B and AN/TKQ-2B shall be examined for the defects listed in MIL-STD-252. In addition, the system shall be visually inspected to determine conformance to the requirements of 3.7.

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4.14 Quality conformance inspection of packaging. Quality conformance inspection of packaging shall be in accordance with group A, and, when required (see 6.2c), the rough handling test of MIL-P-116. Inspection lots shall be in accordance with MIL-P-116.

5. PACKAGING

5.1 Packaging requirements.

5.1.1 AN/AKT-18B. The requirements for packaging shall be in accordance with SPI 1G00248, levels A and B.

5.1.2 AN/TKQ-2B. The requirements for packaging shall be in accordance with MIL-S-55507, levels A and B.

6. NOTES

6.1 Intended use. The Transmitting Set, Radar Data AN/AKT-18B (data transmitting set) is an airborne system which accumulates radar video data and navigational data, encodes the data into a serial digital bit stream, and transmits it to Receiving Set, Radar Data AN/TKQ-2B (data receiving set). Video and navigational information to be transmitted is derived from Radar Surveillance Set AN/APS-94F (radar set). The radar set and data transmitting and receiving sets are components of Radar Surveillance System AN/UPD-7 (system). This system permits fixed target and moving target radar imagery from the airborne radar set to be recorded in real time by ground receiving sets for immediate use by field commanders. The direct support test equipment shall be used in maintenance of the system. The control head controls airborne portion of the data link. The digital encoder receives the FT and MT data and compresses and encodes the data into suitable form for modulation of the FM/FSK transmitter which is contained in the airborne AN/ARC-164(V)16 Radio Set. The radio set transmits FM-FSK data on any one of 7000 channels in the 225 to 399.975 MHz RF spectrum. The operating frequency is selected by the remotely located radio set control. The airborne radio set transmits a continuous FSK signal with 30 watts RF output. The transmitter transmits the data over a UHF link to the ground station where it is received by the receiver of the ground station AN/ARC-164(V)16 Radio Set. A serial digital stream of data is then coupled to the decoder where the data is decoded into discrete digital data words. These words are further processed into either digital or analog signals suitable to drive the RO-495 Recorder-Processor Viewer.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification and any amendment thereto.

b. Level A, level B preservation and packing (see section 5).

c. When rough handling test is required (see section 4).

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d. The contractor should apply for nomenclature in accordance with the applicable clause in the contract.

e. Marking and shipping of samples.

f. Place of final inspection.

g. Number of first articles to be submitted for approval (see 3.1).

6.3 Environmental. Environmental pollution prevention measures are contained in the packaging material specifications referenced herein. Refer to material specifications or preparing activity for recommended disposability methods.

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Preparing Activity:
Army - ER
Project A024

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-T-49438(ER)	Transmitting Set, Radar Data AN/AKT-18B and Receiving Set, Radar Data AN/TKQ-2B (system of)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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